Introduction to Program Synthesis (SS 2026)

Chapter 1 - Introduction

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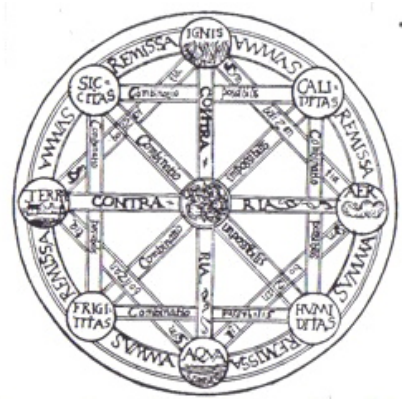
- Leibniz (1679) had the ideal of the following
  - $\rightsquigarrow$ Creation of a *universal language* in which all possible problems can be formulated $\rightsquigarrow$ **Characteristica universalis**
- Church (1930s) developed a formal system called the **lambda calculus**
  - $\rightsquigarrow$ Definition of the notion of computable function in the framework of a calculus
- Turing (1936/7) developed a class of machines that later became known as **Turing machines**
  - $\rightsquigarrow$ Defined the notion of computable function via these machines



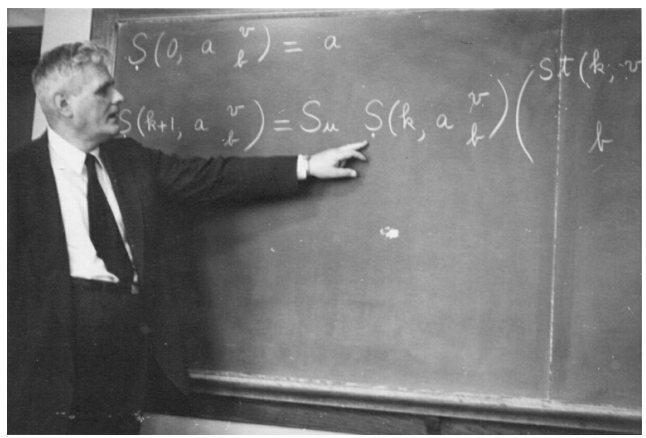
- Calculus (dt. Kalkül) $\rightarrow$ formal system of rules
- Used to derive statements from given statements (axioms)
- A calculus consists of
  - $\rightsquigarrow$ **Building blocks**
    - Alphabet $\rightarrow$ symbols, connectives, structuring signs, ...
  - $\rightsquigarrow$ **Formation rules**
    - Define how building blocks form complex objects or well-formed formulas
    - Analogy to natural language $\rightarrow$ "grammar" of the calculus
  - $\rightsquigarrow$ **Transformation rules**
    - Derivation and deduction rules
    - Transformation to create new objects from them
  - $\rightsquigarrow$ **Axioms**
    - Objects or expressions
    - Formed according to the formation rules of the calculus
    - Obvious principle

Example (Chess Calculus)

Chess game with pieces (axioms) and moves (transformation rules).

- **Formal framework** that can be used in mathematics, logic and programming
- Approach to **systematic solving** problems in certain domain
- Design of suitable **logical frameworks** for **programming languages**
- Examples:
  - **Mathematical** $\rightarrow$ arithmetics, O-Calculus, stochastic calculus
  - **Logic** $\rightarrow$ propositional calculus
  - **Computation** $\rightarrow$ λ-Calculus, Turing machine, Plankalkül

- λ-Calculus $\rightarrow$ minimal model of computation
  - $\rightsquigarrow$ Smallest universal programming language
  - $\rightsquigarrow$ Simple yet powerful system $\rightarrow$ Turing complete $\rightarrow$ universal model of computation
  - $\rightsquigarrow$ Any computable function can be expressed and evaluated
- Developed by Alonzo Church in the 1930's
  - Published in 1941 [Chu85]
  - Study of functional computing
- Introduction of a functional notation: $\lambda x.y$
  - $\rightsquigarrow$ Contemporary notation analogy: $x \mapsto y$
  - $\rightsquigarrow$ *Formalisation* of mathematical functions
    - $\mathrm{f(x)} = x^2 \quad \rightsquigarrow \quad x \mapsto x^2,\ \lambda x.x^2$
    - $\mathrm{f(x,y)} = x^2 + y^2 \quad \rightsquigarrow \quad (x, y) \mapsto x^2 + y^2,\ \lambda x.\lambda y.x^2 + y^2$



- Functions are considered expressions E
  - $\rightsquigarrow$ Parenthesis can be used for clarity $E \Leftrightarrow (E)$
  - $\rightsquigarrow$ Keywords are only λ and the dot
- **Function creation:** Function denotation that has a formal argument x and a functional body $E \rightarrow \lambda x.E$
- **Function application:** Denotation of the application of a function E_1 to the argument $E_2 \rightarrow E_1.E_2$
  - $\rightsquigarrow$ Adopts the convention that function application associates from the left
  - $\rightsquigarrow$ $E_1 E_2 E_3 \ldots E_n \rightarrow (\ldots((E_1 E_2)E_3)\ldots E_n)$
- Syntax of lambda calculus:

```
<expression> := <name> | <function> | <application>
<function> := λ <name>.<expression>
<application> := <expression><expression>
```

- **Abstraction** $\rightarrow$ anonymous functions
- Single transformation rule $\rightarrow$ **variable substitution**
- Single function definition scheme $\rightarrow \lambda x.x$
  - $\rightsquigarrow$ λ symbol $\rightarrow$ start of a function expression
  - $\rightsquigarrow$ Name after $\lambda \rightarrow$ identifier of the function argument
  - $\rightsquigarrow$ Expression after the point $\rightarrow$ function body
- Functions can be applied to expressions $\rightarrow (\lambda x.x)y$
  - $\rightsquigarrow$ Evaluation $\rightarrow$ substitution of the argument x
  - $\rightsquigarrow$ $(\lambda x.x)y = [y/x]\,x = y$
  - $\rightsquigarrow$ $[y/x] \rightarrow$ notation to indicate the substitution of x by y

- Variables can be either free or bound (like in math):
- **Free variable** $\rightarrow$ symbol in an expression that can be substituted
  - $\rightsquigarrow$ Not bound in an abstraction
- **Bound variable** $\rightarrow$ occurrence in the body of the definition is preceded by λ
  - $\rightsquigarrow$ Symbol bound to logical quantifiers or variable-binding operators i.e. $\sum_{x=0}^{N}$ $\prod_{x=0}^{\infty}$ $\forall x$ $\exists x$
- $(\lambda x.xy) \rightarrow$ variable x is bound and y is free
- $(\lambda x.x)(\lambda y.yx)$
  - $\rightsquigarrow$ x in the first expression is bound to the first λ
  - $\rightsquigarrow$ y in the second expression is bound to the second λ
  - $\rightsquigarrow$ x in the second expression is free

- Functions in standard λ calculus are anonymous
  - $\rightsquigarrow$ However, capital letters are commonly used to simplify the notation
  - $\rightsquigarrow$ For instance, the identity function $\mathtt{I}$ serves as a synonym for $\mathtt{I} \equiv (\lambda x.x)$
- **Substitution** $\rightarrow$ fundamental mechanism in λ calculus
- Computational approach to function composition
  - $\rightsquigarrow$ $g(x) := (u \circ v)(x) = u(v(x))$

Example (Identity function)

- We apply the identity function to itself which is an application:
  $\rightsquigarrow$ $\mathtt{II} \equiv \mathtt{I}_1.\mathtt{I}_2 \equiv (\lambda x.x)(\lambda x.x)$
- We can rewrite the expression as:
  $\rightsquigarrow$ $\mathtt{II} \equiv (\lambda x.x)(\lambda z.z)$
- The identity function when applied to itself leads therefore to:
  $\rightsquigarrow$ $\mathtt{II} \equiv (\lambda x.x)(\lambda z.z) = [\lambda z.z/x]\, x = \lambda z.z \equiv \mathtt{I}$

- Outermost parentheses can be omitted
  - $\rightsquigarrow (\lambda x.x) \equiv \lambda x.x$
- Function application to arguments is generally left associated
  - $\rightsquigarrow \underbrace{(\lambda x.x)}_{\text{Function}} \quad \underbrace{1}_{\text{Argument}} \equiv \lambda x.x \quad 1$
- There are various notation for substitution
  - $\rightsquigarrow [y/x]\, x$
  - $\rightsquigarrow \{x \leftarrow y\}$
  - $\rightsquigarrow [x := y]$
  - $\rightsquigarrow [x \mapsto a]$

- λ-calculus knows three rules
- α-**equivalence** or **conversion** $\rightarrow$ renaming of (bound) variables and parameters
  - $\rightsquigarrow$ $\lambda x.(x\,y) \rightarrow_\alpha \lambda x.(x\,z)$
  - $\rightsquigarrow$ $\lambda x.x \equiv \lambda y.y \equiv \lambda z.z$
  - $\rightsquigarrow$ $(\lambda x.x + y) \not\equiv (\lambda y.y + y)$
- β-**reduction** $\rightarrow$ substitution that is performed in the context of application
  - $\rightsquigarrow$ $(\lambda x.t)a \rightarrow_\beta t\,[x \mapsto a]$
  - $\rightsquigarrow$ $(\lambda x.x)\,1 \rightarrow (\lambda 1.1) \rightarrow 1$
- η-**conversion** $\rightarrow$ abstraction over a function
  - $\rightsquigarrow$ $(\lambda x.Mx) \rightarrow_\eta M$
  - $\rightsquigarrow$ x may not a free variable in M
  - $\rightsquigarrow$ Often omitted in λ calculus

- Computing λ functions $\rightarrow$ iterative β-reduction until the normal form is reached which is irreducible
  - $\rightsquigarrow$ $M \rightarrow_\beta M_1 \rightarrow_\beta M_2 \rightarrow_\beta \cdots \rightarrow_\beta N \nrightarrow \beta$
  - $\rightsquigarrow$ β normal form $N \rightarrow$ no longer possible to apply further arguments
  - $\rightsquigarrow$ $\lambda x.x\ \lambda y.y \rightarrow \lambda y.y$

- λ functions have no more than one bound variable and are applied to one argument
  - $\rightsquigarrow$ The following function takes two arguments: $\lambda x.(\lambda y.xy)$
  - $\rightsquigarrow$ The notation can be simplified as: $\lambda xy.xy$
- Arguments are processed from left to right
  - $\rightsquigarrow$ The first argument substitutes the outer λ
  - $\rightsquigarrow$ This process is know as **currying** which means breaking down **a function** that takes **multiple arguments** into **a sequence** of **single argument functions**

$$
\begin{aligned}
\lambda x.(\lambda y.xy) \quad & 12 \\
[x := 1] & \\
\lambda 1.(\lambda y.1y) \quad & 2 \\
[y := 2] & \\
\lambda 1.(\lambda 2.12) & \\
12 &
\end{aligned}
$$

- Arithmetic calculations $\rightarrow$ integral part of a programming language
- We can define a successor function Γ:
  - $\rightsquigarrow \Gamma(0) = 1$
  - $\rightsquigarrow (\Gamma \circ \Gamma) = \Gamma(\Gamma(0)) = 2$
- **Church numerals** $\rightarrow$ Representation of natural numbers
  - $\rightsquigarrow$ Based on n-fold composition function $f^{\circ n} = \underbrace{f \circ f \circ \cdots \circ f}_{n \text{ times}}$
- With λ-calculus we obtain the following numerals:
  - $\rightsquigarrow 0 \equiv \lambda s.(\lambda z.z) = \lambda sz.s$
  - $\rightsquigarrow 1 \equiv \lambda sz.s(z)$
  - $\rightsquigarrow 2 \equiv \lambda sz.s(s(z))$
  - $\rightsquigarrow 3 \equiv \lambda sz.s(s(s(z)))$
- Later more in the framework of the exercise

- Building new terms can be done in two ways:
  - $\rightsquigarrow$ λ-terms $\rightarrow$ build from a variable x and a term M
    - $\lambda x.M$
  - $\rightsquigarrow$ Applications $\rightarrow$ build from two terms M and N
    - Written as $(M\ N)$
- Terms can be considered and represented as trees
  - $\rightsquigarrow$ Inner nodes (non-terminals) are λ terms $\rightarrow$ functions or applications
  - $\rightsquigarrow$ Outer nodes (terminals) are variables

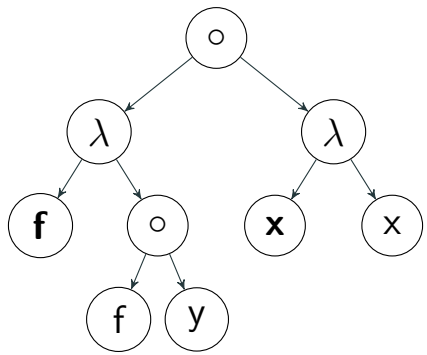


Figure: Tree representation of lambda expression $\Psi = ((\lambda f.(f\ y))\,(\lambda x.x))$

- Pure λ-calculus is a fundamental ingredient of functional programming languages
  - $+$ reduction strategy
  - $+$ data types
  - $+$ type system
- Building blocks of functional programs
  - $\leadsto$ Composition of terms

References I

[Chu85] A. Church. *The Calculi of Lambda-Conversion*. Princeton University Press, 1985.